RQ2 - Source and Platform Bias in GlobalGeoTree

How data sources shape geographic and taxonomic patterns

Research Question

RQ2 - Source and Platform Bias
To what extent do data sources shape the observed geographic and taxonomic patterns in **GlobalGeoTree**?

This report asks:

1. Which sources contribute the largest share of observations?
2. Is the dataset dominated by citizen-science platforms such as iNaturalist?
3. Do different sources cover different countries or regions?
4. Are some sources associated with specific years, countries, or species groups?

The analysis is descriptive. Strong source-specific patterns should be interpreted as evidence of **recording and platform bias**, not only as biological patterns in tree distributions.

How to read the visualisations. Each figure is included for a specific reason: first to show whether a few sources dominate the dataset, then to check whether that dominance changes by country, year, and taxonomic group. The short interpretation after each figure explains what the pattern means and why it matters for answering RQ2.

Data Loading

Measure	Value
Observation file	GlobalGeoTree.csv
Rows analysed	6,263,345
Distinct sources	2,493
Distinct source groups	5
Distinct countries	221
Distinct species	21,001
Year column available	Yes
Coordinate columns available	Yes
Category source	GlobalGeoTree-10kEval-900.csv joined by species_key
Observations with known category	2,205,871
Species with known category	941

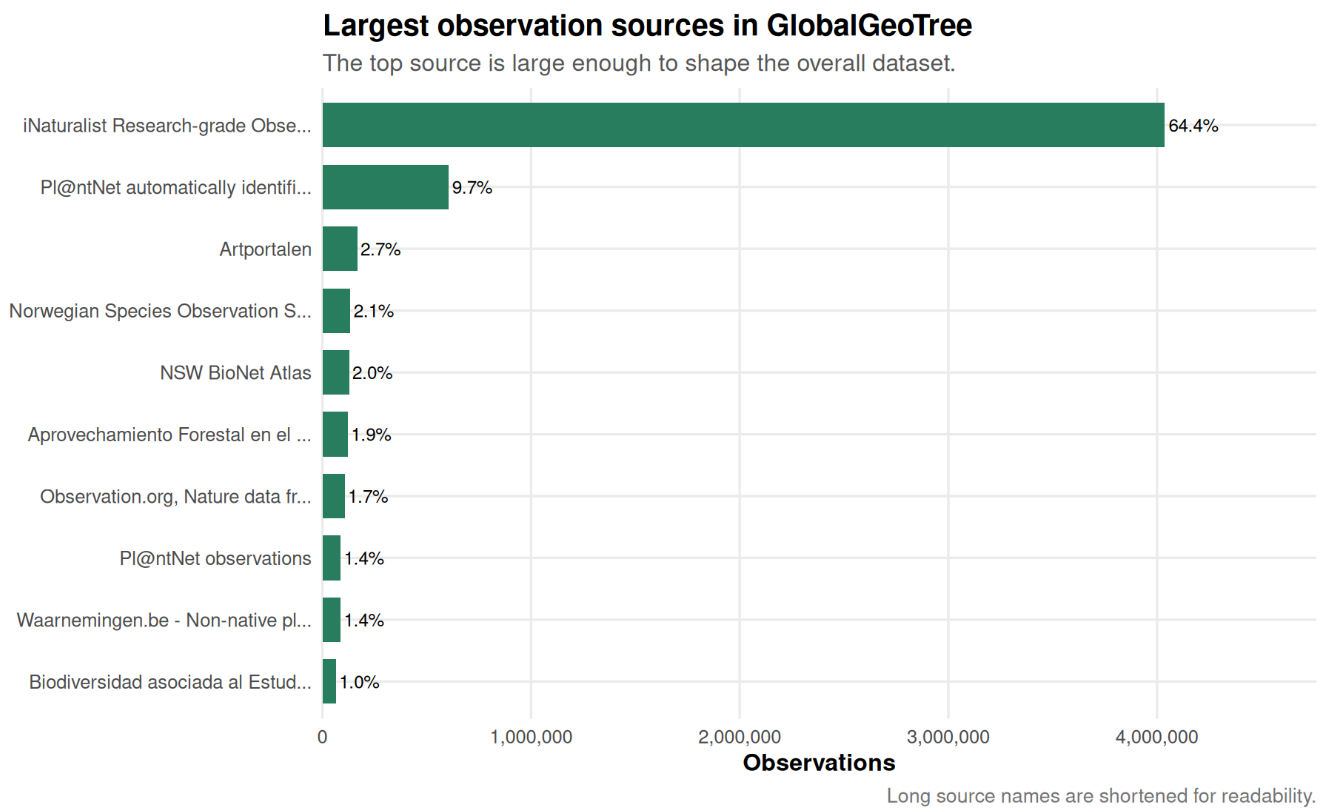
This report uses 6,263,345 observations.

Headline Result

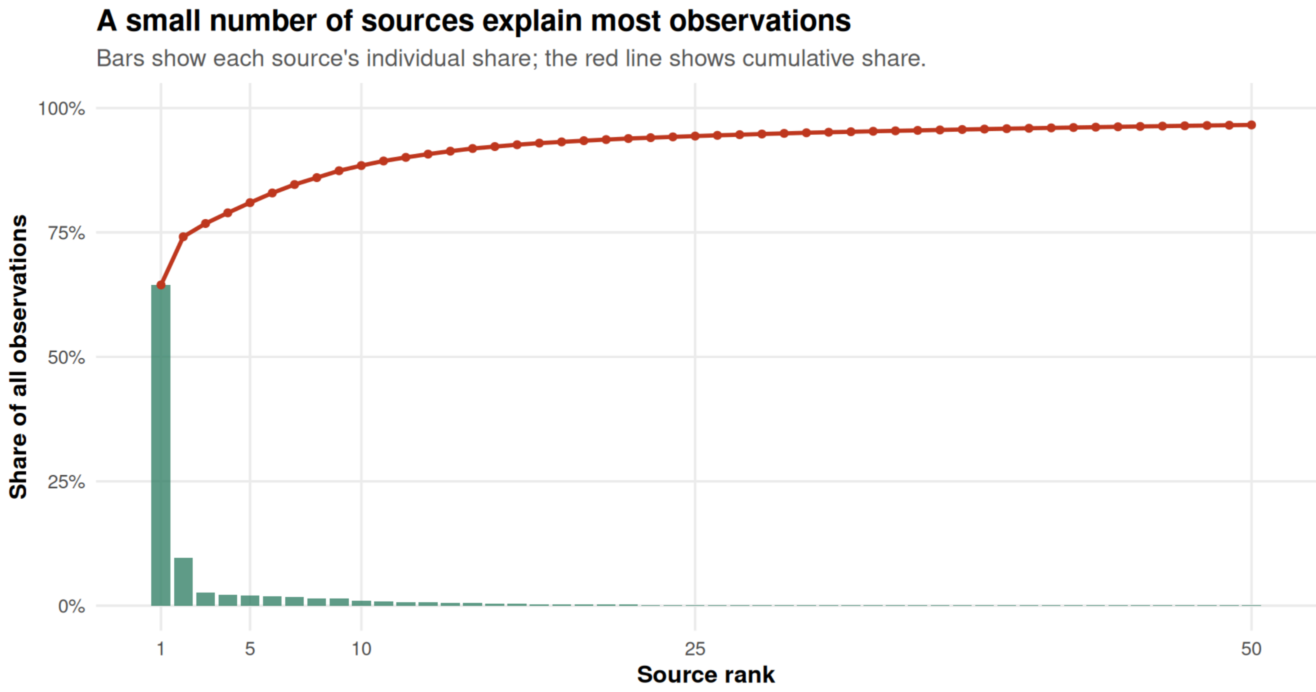
The source structure is highly concentrated: **iNaturalist alone contributes 64.5%** of observations, while **iNaturalist and Pl@ntNet together contribute 75.5%**. Therefore, many observed geographic, temporal, and taxonomic patterns should be read partly as platform-coverage patterns.

Source Concentration

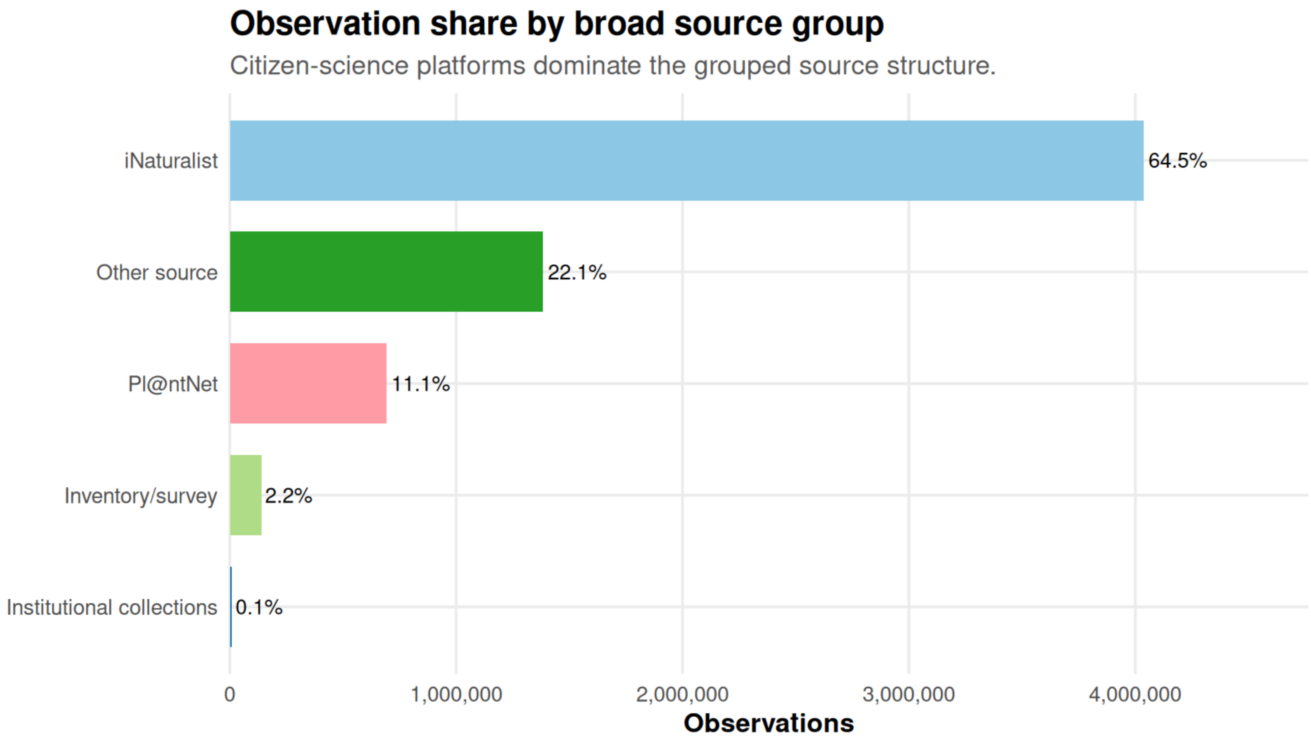
Why this visualisation is needed. Before comparing countries, years, or species, we first need to know whether the dataset is balanced across sources. If one source is extremely large, it can shape the whole dataset.



Why this visualisation is needed. The Pareto chart shows concentration more clearly than a normal table. The bars show individual source shares, while the line shows how quickly the cumulative share rises.



Why this visualisation is needed. Individual source names are detailed, but broad groups are easier to discuss. This plot shows whether citizen-science platforms dominate after grouping sources into wider categories.



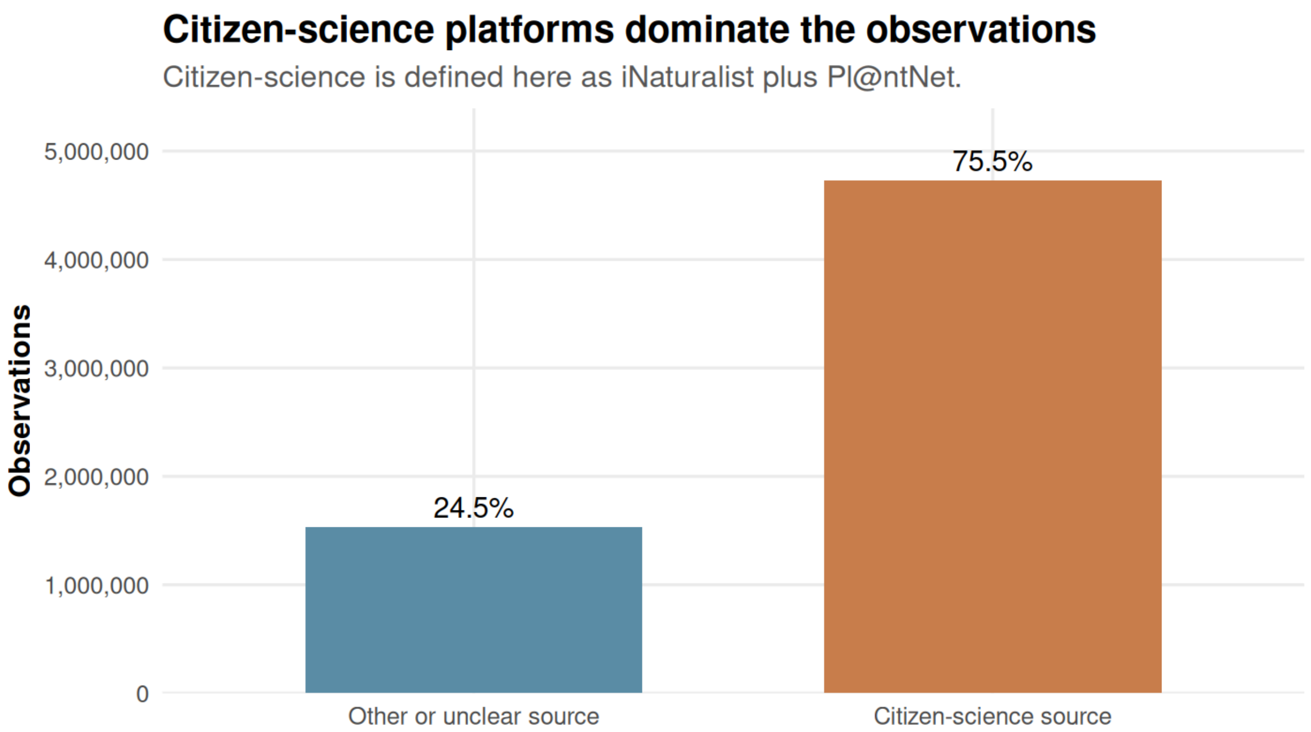
Source	Observations	Share	Cumulative share
iNaturalist Research-grade Observations	4,036,465	64.4%	64.4%
Pl@ntNet automatically identified occurrences	606,258	9.7%	74.1%
Artportalen	166,929	2.7%	76.8%
Norwegian Species Observation Service	134,097	2.1%	78.9%

Source	Observations	Share	Cumulative share
NSW BioNet Atlas	128,009	2.0%	81.0%
Aprovechamiento Forestal en el área de influencia del Proyecto Hidroeléctrico Ituango	121,900	1.9%	82.9%
Observation.org, Nature data from around the World	106,352	1.7%	84.6%
Pl@ntNet observations	86,806	1.4%	86.0%
Waarnemingen.be - Non-native plant occurrences in Flanders and the Brussels Capital Region, Belgium	86,599	1.4%	87.4%
Biodiversidad asociada al Estudio de Impacto Ambiental del proyecto: Interconexión Cuestecitas - Copey - Fundación 500/220 mil voltios	65,000	1.0%	88.4%

Interpretation. The distribution is not balanced across sources. A few large platforms dominate the evidence base, so global summaries are strongly affected by the behaviour, geography, and taxonomy covered by those platforms.

Citizen-Science Dominance

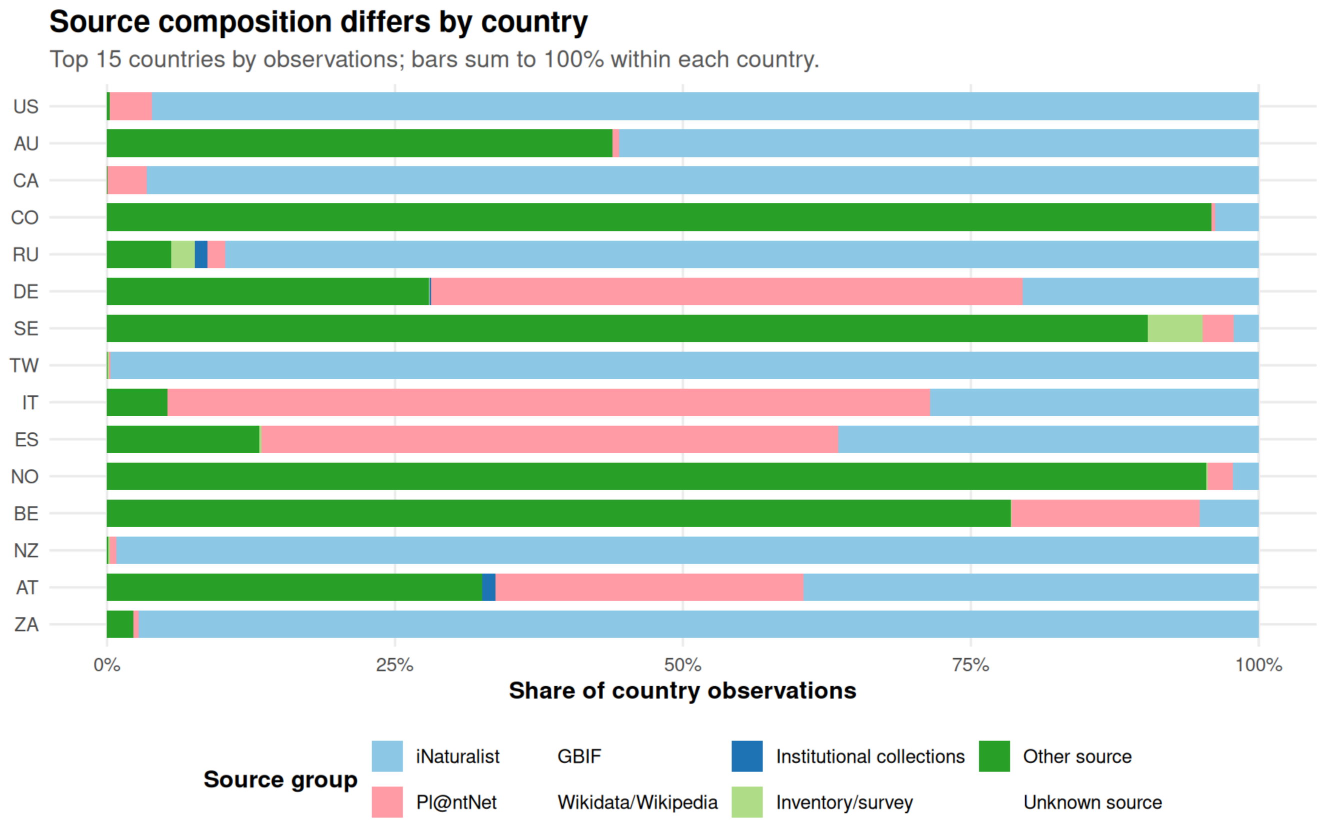
Why this visualisation is needed. The research question specifically asks whether GlobalGeoTree is dominated by citizen-science platforms. This simple two-bar chart directly compares citizen-science records with all other or unclear sources.



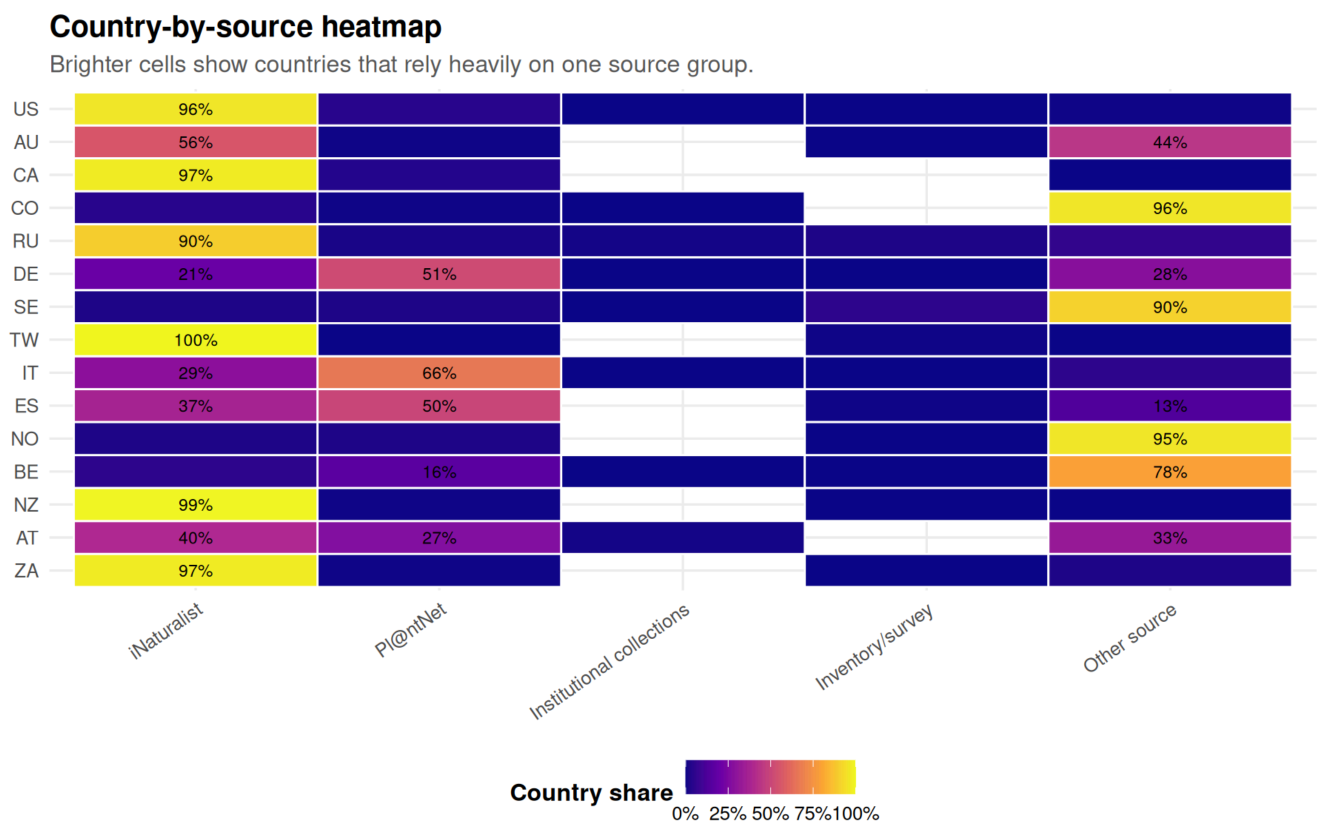
Interpretation. Citizen-science dominance does not make the data unusable, but it changes how the data should be interpreted. Places and species that are popular with users or easy to photograph may be overrepresented.

Countries and Regional Coverage

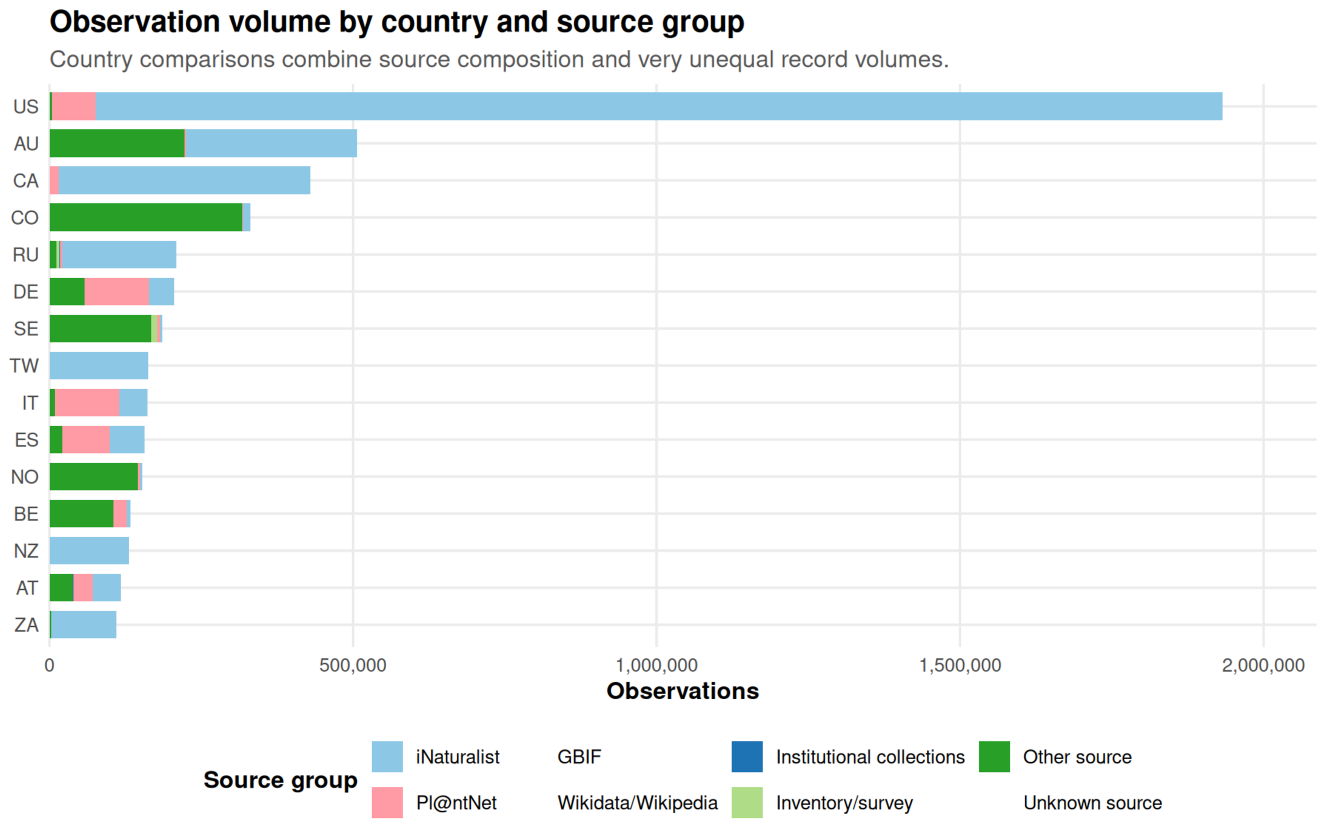
Why these visualisations are needed. A global source pattern can hide strong differences between countries. These plots check whether countries are represented by similar source mixtures or whether each country depends on different platforms.



Why this visualisation is needed. The stacked bars show the full composition, but the heatmap makes extreme dependence on one source easier to spot.



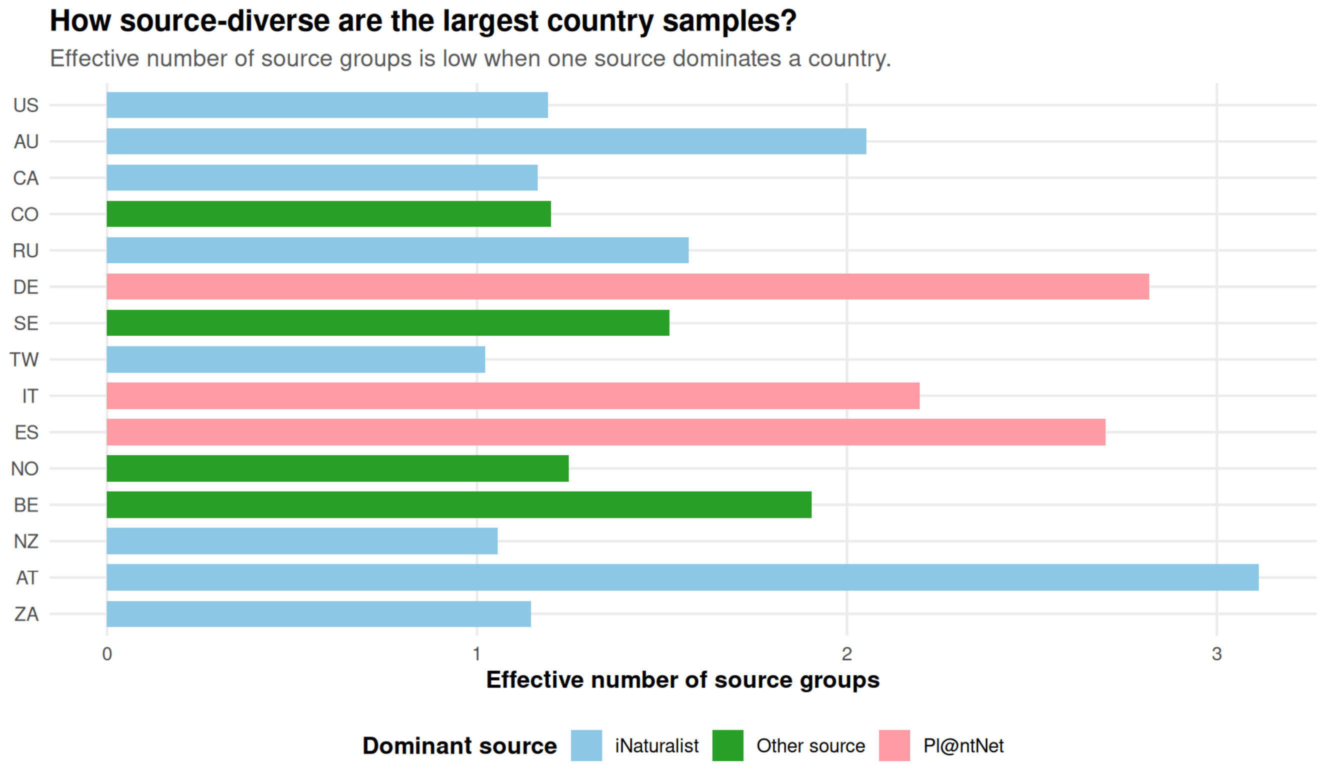
Why this visualisation is needed. The previous plot uses percentages, but percentages can hide the actual amount of data. This count plot shows whether countries have similar source mixes and similar record volumes.



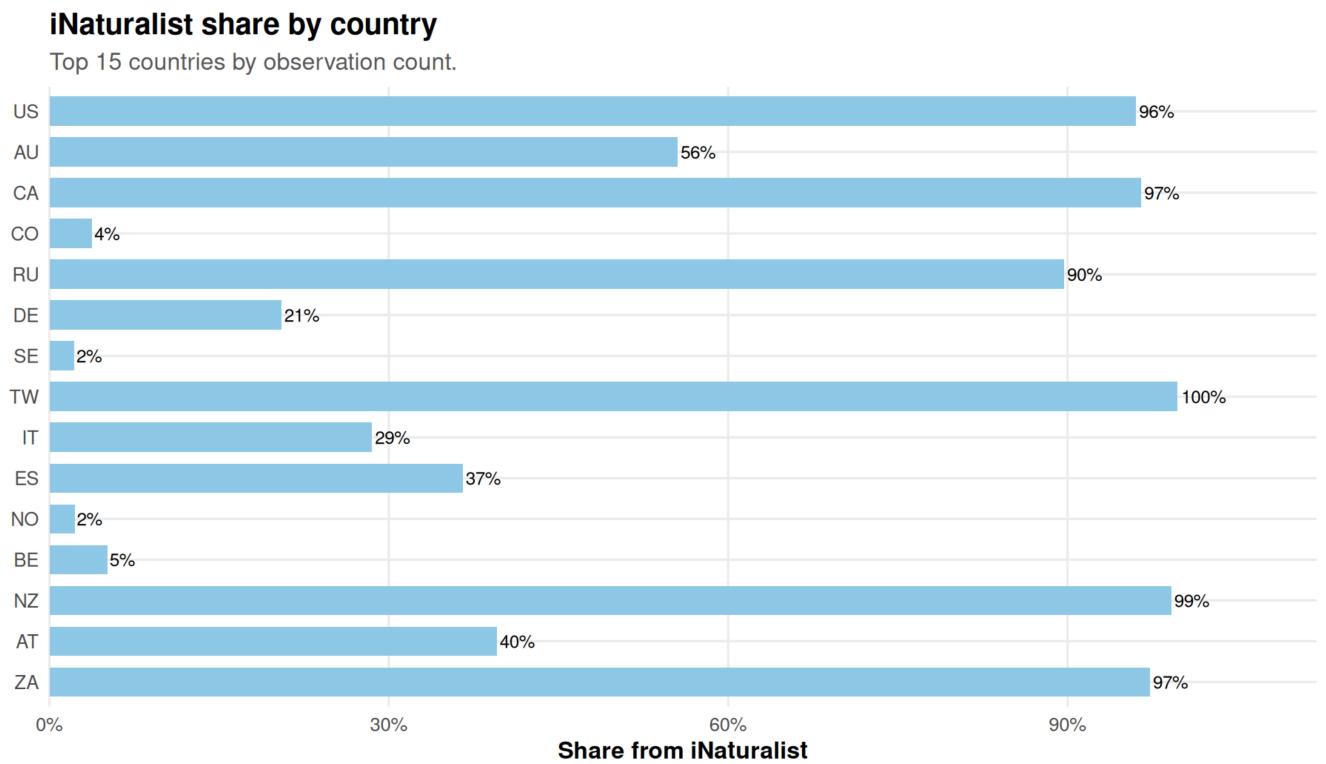
Interpretation. Countries differ both in the number of observations and in the source mix. This means that geographic comparisons can be biased by platform adoption. A country dominated by iNaturalist is not directly comparable to a country dominated by local inventories or unknown sources.

Source Diversity by Country

Why this visualisation is needed. Source diversity gives a single summary number for each country. A low value means one source dominates; a higher value means observations are spread across several source groups.



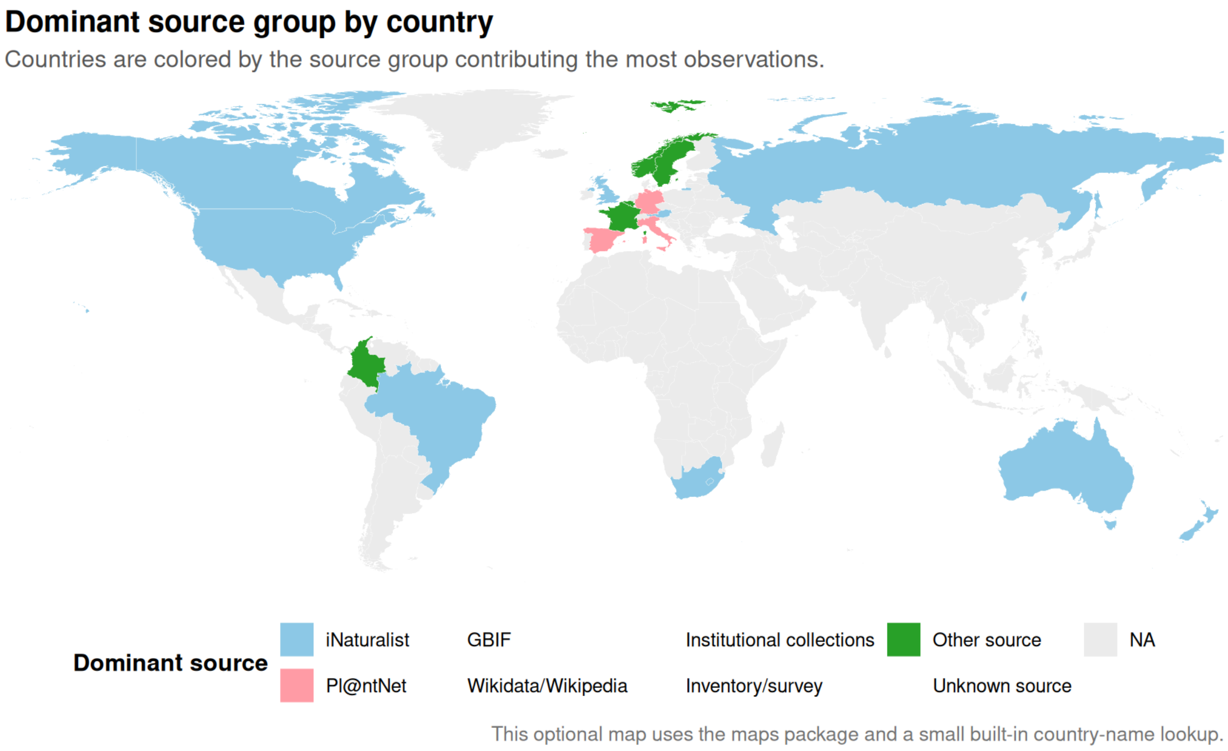
Why this visualisation is needed. Since iNaturalist is the largest platform overall, this chart checks whether that dominance is global or concentrated in particular countries.



Interpretation. Source diversity provides a compact way to see which countries depend on one platform and which countries combine several source groups. Low diversity means higher risk that country-level patterns are platform-specific.

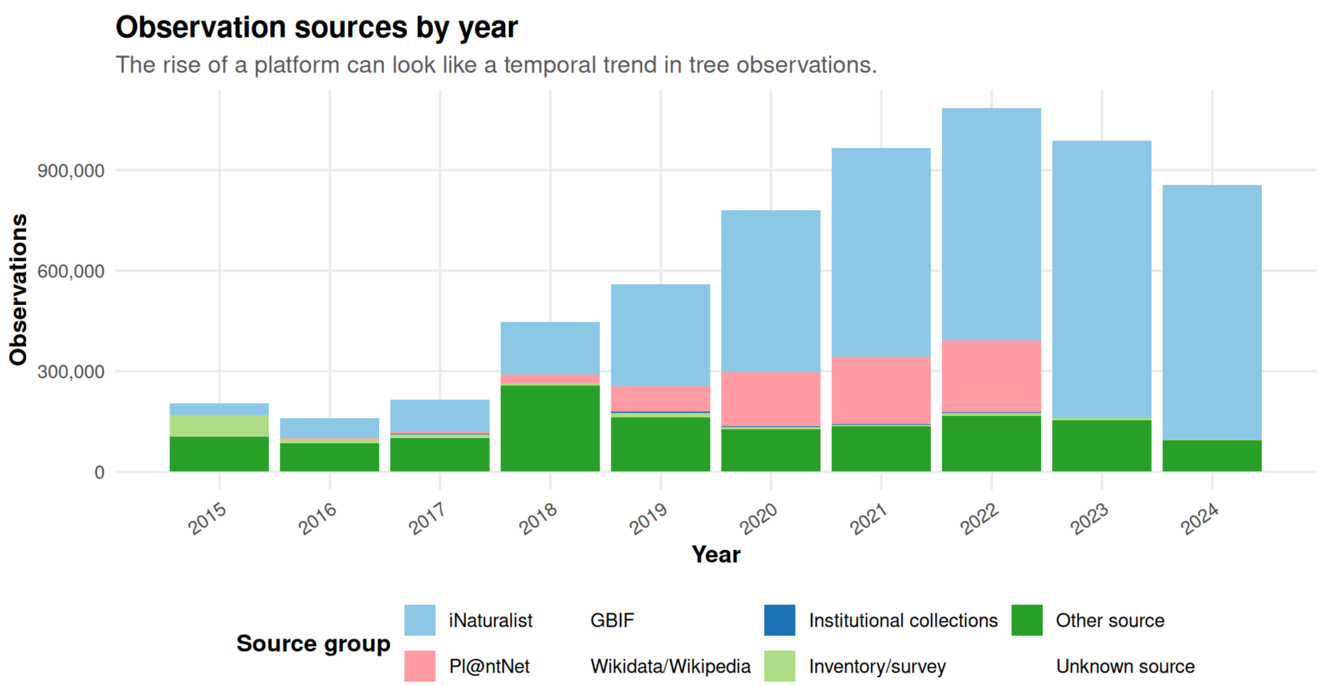
Optional Map: Dominant Source by Country

Why this visualisation is needed. The map gives a quick spatial overview of source dominance. It should be read as a source-coverage map, not as a biological distribution map.

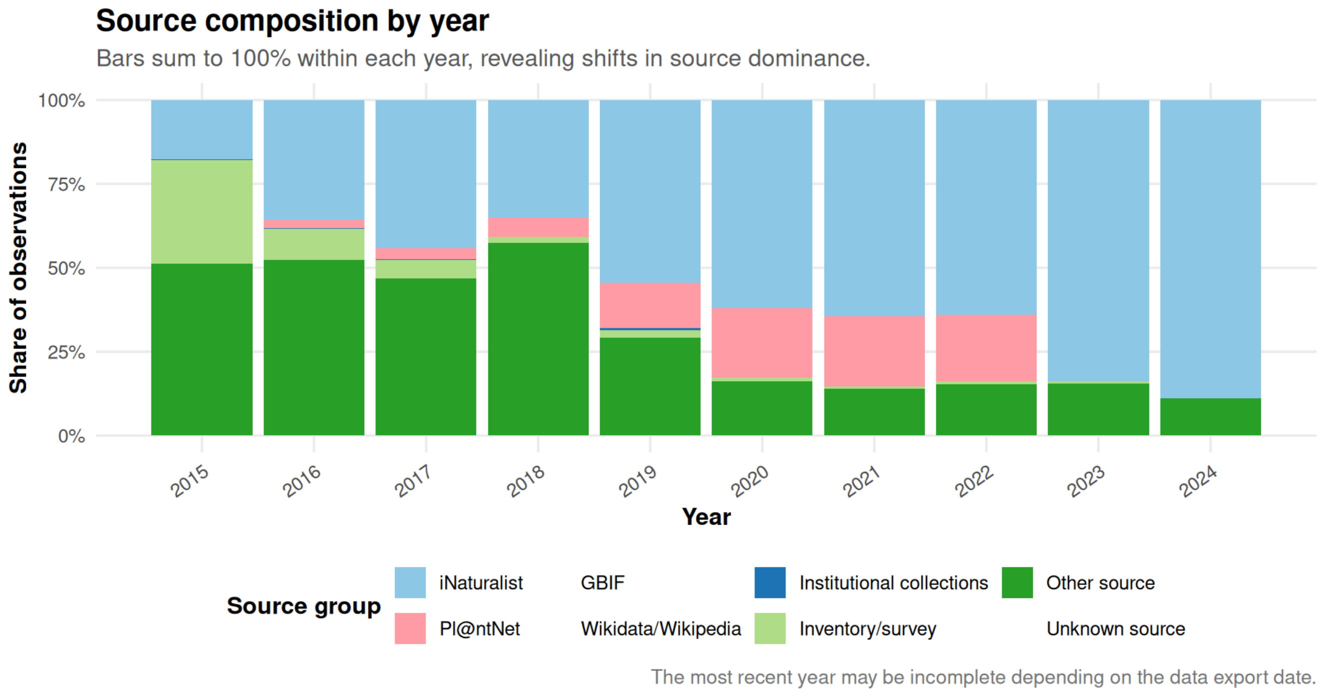


Sources Over Time

Why these visualisations are needed. Time trends can be misleading if a platform became popular recently. These plots check whether changes over time are partly changes in source coverage rather than changes in tree occurrence.



Why this visualisation is needed. The count plot shows volume, while this percentage plot shows whether the source composition changed within each year.

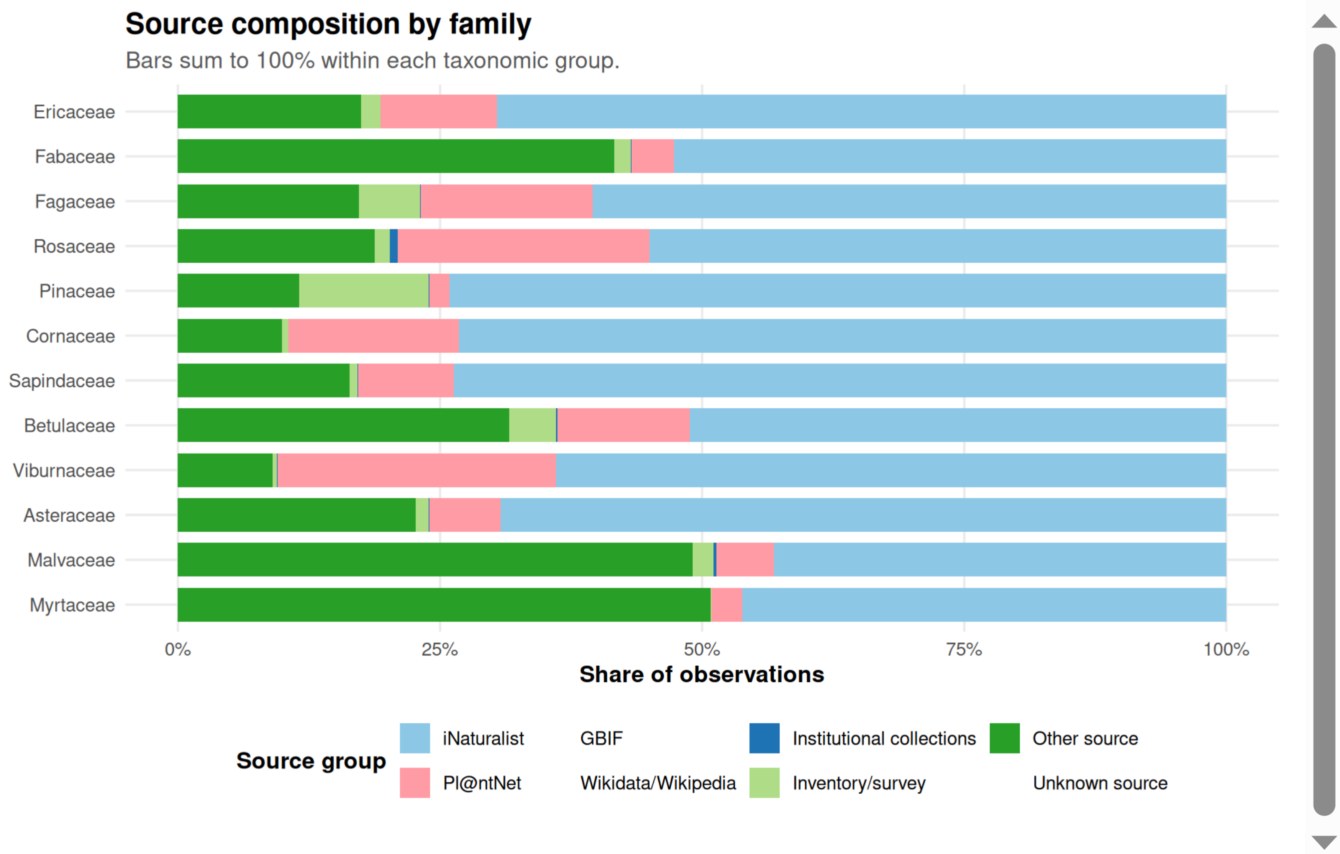


Interpretation. Temporal patterns are not source-neutral. If iNaturalist or Pl@ntNet grows in recent years, the dataset may show an apparent rise in observations even if actual tree occurrence is unchanged.

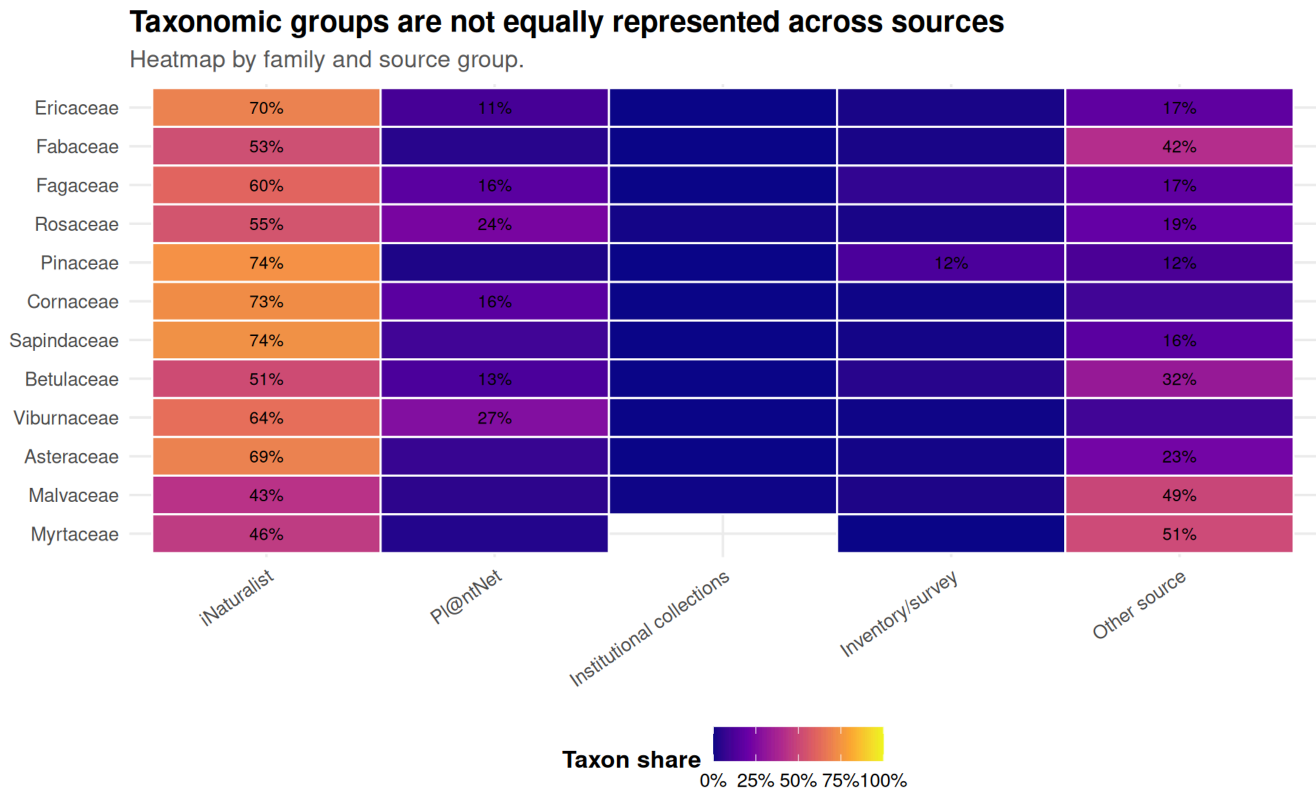
Sources and Taxonomic Groups

Why these visualisations are needed. Source bias can also be taxonomic. For example, some platforms may record ornamental or easy-to-photograph families more often than other groups.

This section checks whether source groups disproportionately record particular families, genera, or species. Family is used first when available because it is more readable than hundreds of species names.



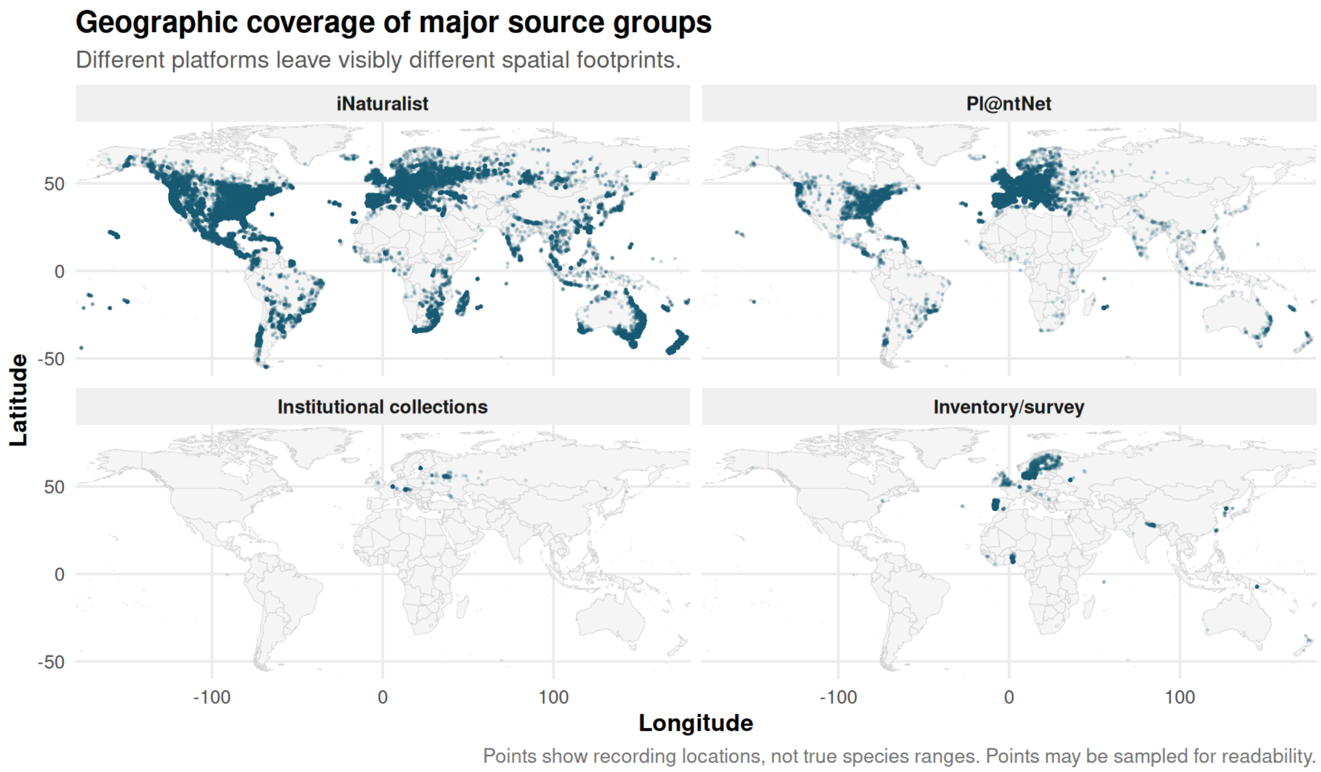
Why this visualisation is needed. The heatmap is a second view of the same taxonomic pattern. It makes unusually high source shares easier to see than stacked bars alone.



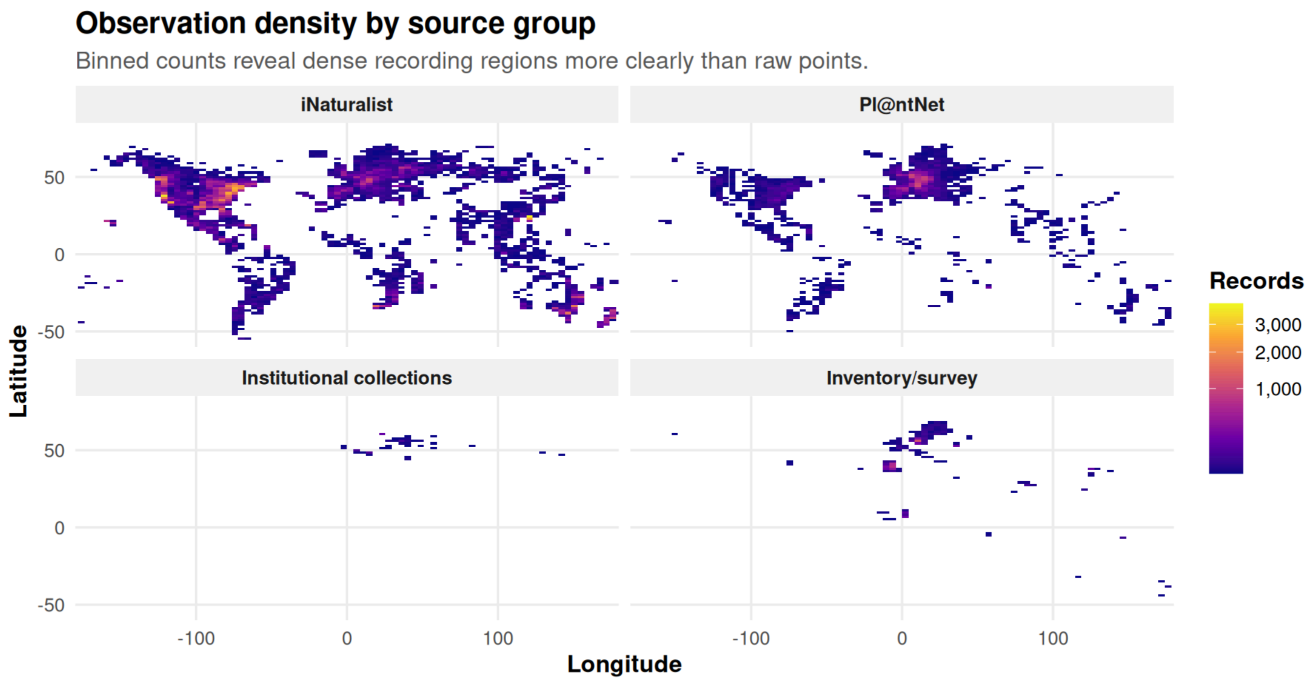
Interpretation. If a family or genus is mostly recorded by one platform, its geographic pattern may partly reflect that platform's user base and coverage.

Geographic Footprints of Major Sources

Why these visualisations are needed. The maps show where the main source groups actually record data. This is important because spatial clusters may reflect user activity and platform coverage, not only real tree distributions.



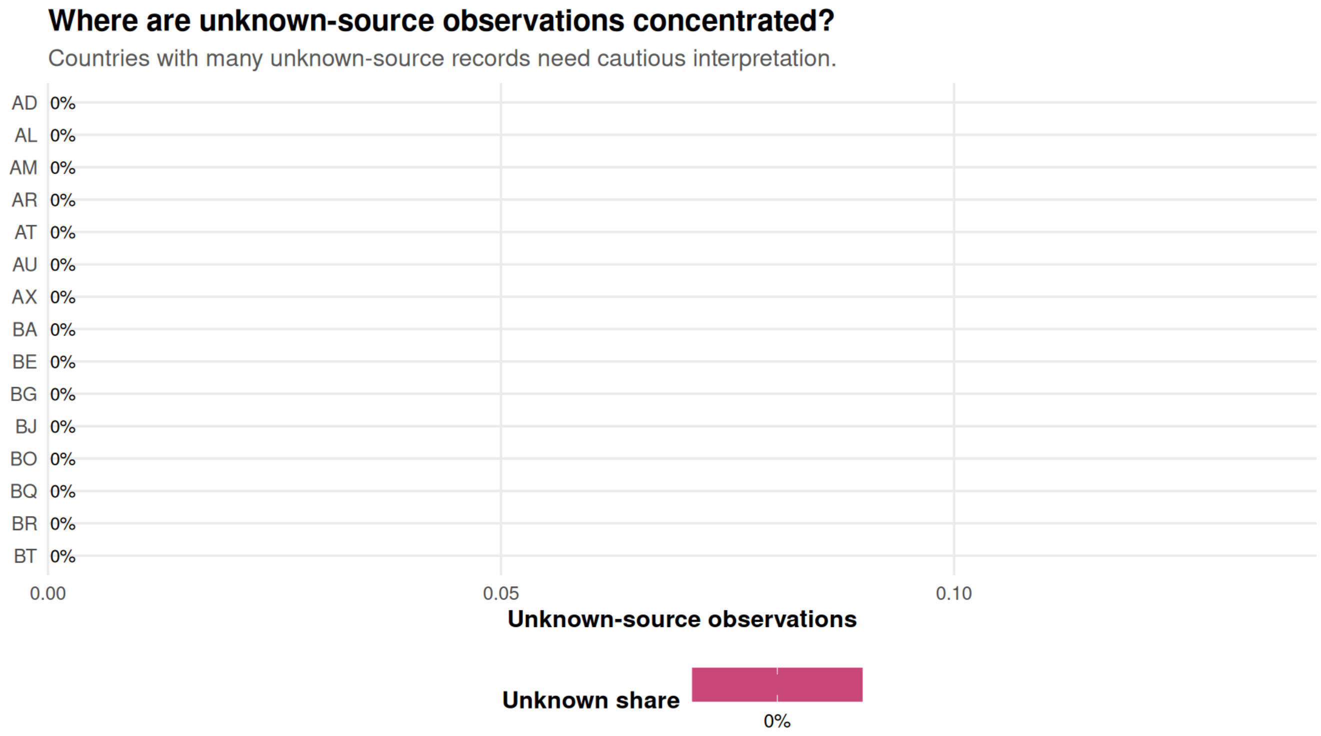
Why this visualisation is needed. Raw points can hide dense clusters because many records overlap. Binning the points makes high-record areas easier to compare across source groups.



Interpretation. The source maps show that platform coverage is geographically uneven. Some regions are visible mainly because they are covered by a particular platform or source pipeline.

Unknown Source Diagnostics

Why this visualisation is needed. Unknown sources are not just missing labels; they reduce confidence in source-specific interpretation. This plot shows whether unknown-source records are concentrated in particular countries.



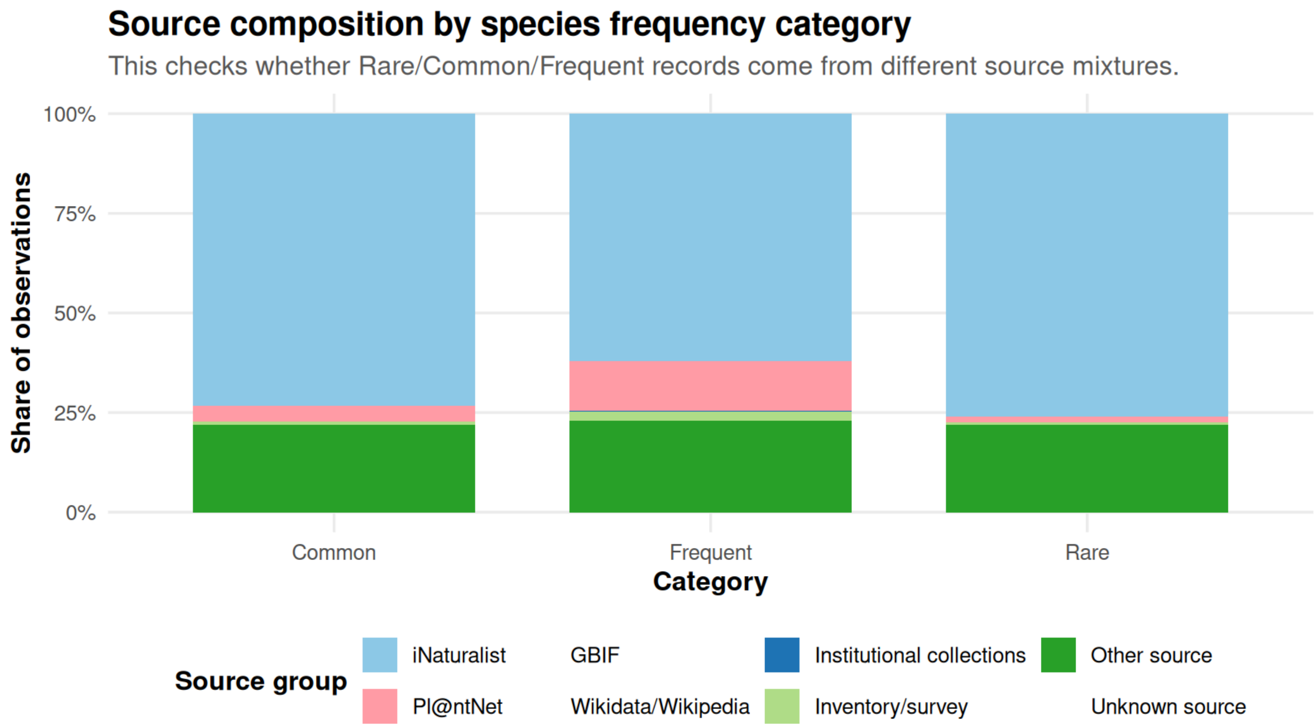
Interpretation. Unknown source records are a metadata-quality issue. If they are concentrated in particular countries, those countries have weaker evidence for source-specific interpretation.

Rare, Common, and Frequent Categories

Why this visualisation is needed. If Rare, Common, and Frequent categories come from different source mixtures, then comparisons between these categories may partly reflect data-source bias.

This section is shown only when category labels are available either inside **GlobalGeoTree** or through the optional eval lookup file.

Measure	Value
Category source	GlobalGeoTree-10kEval-900.csv joined by species_key
Observations with known category	2,205,871
Species with known category	941



Interpretation. If the bars are very similar, the category comparison is less affected by source mix. If one category has much more iNaturalist, Pl@ntNet, or unknown-source data, then category-level findings should be interpreted with more caution.

Overall Interpretation

This analysis shows that GlobalGeoTree is not source-neutral. The dataset contains millions of observations from thousands of named sources, but the observation structure is dominated by a small number of platforms, especially iNaturalist and Pl@ntNet. Source composition also varies by country, year, and taxonomic group. Therefore, observed geographic and taxonomic patterns should be interpreted as a combination of tree occurrence, recording effort, platform popularity, and source-specific data integration.

For downstream analyses, I would report source composition next to any global, country-level, temporal, or taxonomic result. Where possible, I would also repeat key analyses with and without the largest citizen-science source to test robustness.

Main limitation. These figures describe patterns in the available records. They do not prove that one source is biased in a negative way; they show where source coverage is uneven and where interpretation needs extra care.